E1.07 – Power Rangers

Statement of Work

Avash Aryal - Project Manager (PM)

Ashok Paudel

Alok Parajuli

Michelle Su

Sponsored by:

TXST (Dr. Dinusha Herath Mudiyanselage)

601 University Dr

San Marcos, TX, 78666

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Version	Summary of Changes	Date
V0.1	Initial team formation; project scope and subsystem assignments drafted	08/31/26
V0.2	Executive Summary first draft completed; team roles finalized	09/03/26
V0.3	Top-Level Block Diagram created; subsystem inputs and outputs defined	09/07/26
V0.4	Product Features sections drafted for all five features	09/12/26
V0.5	Key Milestones table populated; D1 and D2 timeline established	09/15/26
V1.0	First complete draft of SOW submitted for internal review	09/21/26
V1.1	Executive Summary revised to meet 300-word limit; citation added	09/22/26
V1.2	Author table corrected, reviewed and formatted document to align with SOW documentation recommendation	09/23/26

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1. Executive Summary

The Executive Summary was written by Avash Aryal and Michelle Su.

Our product is a hybrid solar and wind DC power conversion system that harvests renewable energy from a 12 W solar array and a belt-driven wind generation stage using four R280 DC motors as generators. Both sources are combined onto a common DC bus and regulated by a GaN boost converter operating in constant-current/constant-voltage (CC/CV) mode to charge a 12.8 V, 5 Ah LiFePO4 battery. A downstream DC-DC buck converter reduces the battery voltage to a regulated 5 V bus for model loads. An Arduino Mega 2560, INA226 power monitors, and a 3.5-inch capacitive touchscreen will provide real-time system monitoring.

Wide-bandgap (WBG) semiconductor devices such as GaN HEMTs offer lower switching losses, higher switching frequencies, and reduced heat dissipation than conventional silicon MOSFETs, making them well suited for compact, high-efficiency power electronics [1]. Texas State University's Department of Electrical Engineering is advancing GaN power device research through the UWBG Power Converter initiative. This project supports that effort by designing and characterizing a GaN-switched boost converter using the Renesas TP65H150G4PS SuperGaN HEMT as a testbed for future GaN research at TXST.

Our team will develop and test the power conversion system over two semesters. During D1, we will characterize the solar and wind sources, finalize the system design, order components, build the GaN boost converter, and implement Arduino CC/CV control. We will also integrate the INA226 sensors and touchscreen display and build and test the GaN buck converter before integrating all subsystems. By the end of D1, we will have a working prototype operating from solar, wind, and combined sources. During D2, we will transfer the prototype to a PCB, build the smart city model, and complete system testing. Testing will measure efficiency, voltage ripple, and output regulation under varying loads to verify that the final system meets its design requirements.

Team Member	Subsystem / Primary Responsibility
Avash Aryal	Power electronics design and development, including boost-converter design, gate-drive circuitry, CC/CV battery charging, and boost/buck converter development with Michelle.
Ashok Paudel	Monitoring and control development, including INA226 power monitoring, PWM control, protection, and Arduino/software control development with Alok.
Alok Parajuli	Voltage/current sensing, user-interface development, renewable-source characterization, configuration and testing, and Arduino/software control development with Ashok.
Michelle Su	Energy-storage and load-system design, including battery integration, power budgeting, 5 V distribution, load validation, and boost/buck converter development and testing with Avash.

Table 1: Preliminary Subsystem Responsibilities

2. Top Level Block Diagram

The top-level block diagram was drawn by Ashok Paudel and Alok Parajuli.

The top-level diagram of our system is shown in Figure 1. The system is divided into five main subsystems: wind generation, solar generation, power management and energy storage, monitoring and control, and load distribution. The solar and wind subsystems provide renewable energy input to the CC/CV boost converter, which regulates the power used to charge the LiFePO4 battery. The stored energy is then passed through a buck converter to provide a regulated 5 V DC bus for the model loads. The monitoring and control subsystem measures voltage and current at different points in the system and uses the Arduino to control and monitor the converters and overall system operation.

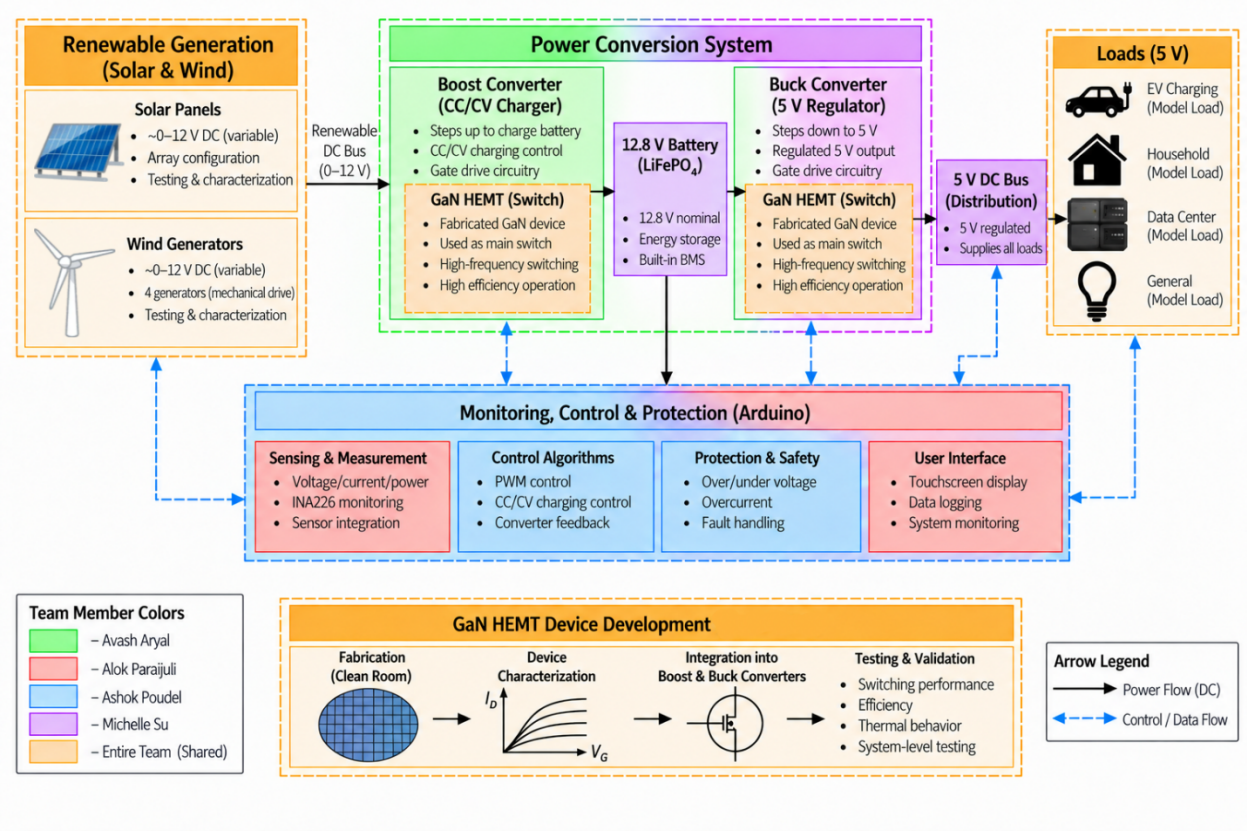


Figure 1: Top-Level Block Diagram

Key inputs to the system are the DC power generated by the solar and wind generation subsystems. The solar array is configured to provide approximately 12 V and up to 1 A under rated conditions, while the four R280 wind generators are designed to provide approximately 2–5 W of combined electrical power. These two renewable power sources are combined and supplied to the power conversion and energy storage subsystem. Key outputs of the system are a regulated 5 V DC supply for the model loads and real-time voltage, current, and power data displayed through the system interface.

3. Product Features

Feature 1: Solar and Wind Energy Harvesting

This section was written by Alok Parajuli.

The system captures DC electrical energy from two independent renewable sources. The solar subsystem combines four 6 V / 3 W / 500 mA monocrystalline panels configured to produce a nominal 12 V / 1 A / 12 W output. The actual output voltage and current will vary based on environmental conditions such as light intensity, panel angle, shading, and temperature, so the boost converter is designed to handle the full range of possible input conditions rather than assuming a fixed 12 V input. The wind subsystem simulates wind energy harvesting using a 12 V DC drive motor as the mechanical power source, which spins a belt connected to four R280 brushed DC motors arranged in a belt-and-pulley configuration. Since the R280 motors are driven mechanically rather than electrically, they operate as generators, each converting rotational motion into a small DC voltage through back-EMF, producing a combined estimated output of 2-5 W depending on drive motor speed and generator loading. An SB540 Schottky diode on each branch blocks reverse current before both sources merge at the common DC bus. Table 2 summarizes key source parameters.

The converter is designed to handle the full range of source conditions listed below.

Parameter	Min	Max	Notes
Solar Array Output Voltage	~9 V	~14 V	Actual Vmp varies with irradiance; design for full range
Solar Array Output Current	0 A	~1.0 A	Expected current 1 A; actual depends on conditions
Wind Generator Combined Output Power	~0 W	~5 W	Depends on drive RPM and load; target 2-5 W

Table 2: Renewable Source Parameters

Feature 2: GaN-Based Boost Converter with CC/CV Battery Charging

This section was written by Avash Aryal.

The GaN-based boost converter serves as the main power conversion stage between renewable energy sources and the battery. Since the voltage generated by the solar and wind subsystems can vary with operating conditions, the converter steps up and regulates the combined DC input to properly charge the 12.8 V $LiFePO_4$ battery. A GaN HEMT is used as the main switching device and is controlled using a PWM signal at approximately 100 kHz. The converter operates in constant-current/constant-voltage (CC/CV) charging mode. During constant-current operation, the charging current is regulated up to 1 A. As the battery approaches the 14.4 V charging limit, the system transitions to constant-voltage operation and allows the charging current to decrease. Battery voltage and charging current are continuously measured and used as feedback to adjust the converter duty cycle. This allows the system to safely charge the battery while adapting to changes in the available renewable power.

Feature 3: Real-Time Monitoring and Display

This section was written by Ashok Paudel.

The Arduino Mega 2560 serves as the central controller for the entire system. It generates a PWM signal to drive the TC4427AEPA gate driver, which switches the GaN HEMT in both boost converter and buck converter. The duty cycle of this PWM signal directly controls the output voltage and current, enabling the CC/CV charging loop for the LiFePO4 battery. The Arduino also polls up to eight INA226 power-monitor modules over I2C, reading bus voltage, shunt current, and computed power at each measurement point across the system. Using these readings, it adjusts the PWM duty cycle in real time to maintain the target charge profile and computes overall system efficiency. All monitored parameters (solar input power, wind input power, bus voltage, battery charging current and voltage, and load power) are streamed to a Hosyond 3.5-inch ST7796U-based touchscreen display over 4-wire SPI at a minimum refresh rate of 1 Hz.

Feature 4: GaN-Based Buck Converter and 5 V Power Regulation

This section was written by Ashok Paudel and Avash Aryal.

The buck converter accepts the 12.8 V battery output and steps it down to a regulated 5 V DC bus to supply the model loads described in Feature 5. Like the boost stage, the switching device is a GaN HEMT, taking advantage of its low switching losses and high switching frequency capability to achieve a compact, high-efficiency design. A dedicated gate driver controls the GaN HEMT switching based on PWM signals from the Arduino Mega, which monitors the output voltage via an INA226 module and adjusts the duty cycle to maintain regulation. The buck converter is designed to supply up to 1 A continuously at 5 V, giving a maximum output power of 5 W. Component selection, switching frequency, and efficiency targets will be finalized in D1.

Feature 5: Energy Storage and 5 V Load Supply

This section was written by Michelle Su.

Harvested renewable energy is stored in a KBT 12.8 V, 5 Ah LiFePO4 battery with built-in BMS (Battery Management System), which handles low-battery cutoff and cell protection. The battery accepts CC/CV charge from the boost converter at 1.0 A CC and a 14.4 V CV limit. A DC-DC buck converter steps the battery voltage down to a regulated 5 V DC bus, which serves as the primary supply rail for all model loads. The load network represents a miniature city model consisting of LED street lighting, a data center represented by a cooling fan, and electric vehicle charging stations represented by load LEDs. Each load branch is individually fused and monitored by an INA226 module, allowing per-branch power consumption to be logged and displayed in real time. Total estimated load current draw is under 1 A at 5 V, giving a worst-case load power of approximately 5 W. The buck converter is sized to supply this load continuously from the battery while the renewable sources are inactive.

4. Key Milestones

Table 3 shows the key project milestones for both D1 and D2 semesters, including course-set checkpoints and project-specific objectives.

Milestone	Date
D1	
Project Team Assignment	08/31/26
Order all project materials	09/08/26
Finalize subsystem assignments; block diagram v1.0 complete	09/10/26
Solar and Wind Source Characterization Complete	09/22/26
SOW submitted	09/23/26
Boost converter PCB schematic and layout complete	10/06/26
Initial Product Specification Complete	10/21/26
Boost converter power stage assembled; open-loop bench test	10/27/26
Buck converter assembled; 5 V output verified	10/27/26
Arduino CC/CV loop verified; closed loop charging confirmed	11/03/26
Wind generation stage mechanical assembly and characterization complete	11/07/26
INA226 sensor array and touchscreen display integration complete	11/10/26
Full-system integration (solar + wind + boost + battery + buck + display)	11/14/26
Subsystem-level bench testing and debugging	11/17/26
Initial Design Review	11/18/26
Full-system performance testing and validation	11/24/26
D1 Senior Design Day	12/03/26
D2	
GaN MOSFET Fabrication complete (boost and buck stages)	02/09/27
Enclosure and city model load network build-out complete	02/23/27
D2 Full Function Demo	TBD
Formal load testing and efficiency campaign across all operating modes	03/29/27
PCB-level performance testing and validation	04/12/27
Final Design Review	TBD
D2 Senior Design Day	TBD

Table 3: Milestone Schedule

5. Approvals

The signatures of the people below indicate an understanding of the purpose and content of this document by those signing it. By signing this document, you indicate that you approve of the proposed project outlined in this Statement of Work, the division of work, the Ground Rules and that the next steps may be taken to create a Product Specification and proceed with the project.

This document is based upon and supersedes the SOW Template V8.2. Deviations (versus clarifications) from the PDR have been clearly noted. For any requirements not listed in this SOW, the PRD requirements shall remain in effect.

Approver Name	Title	Signature	Date
Avash Aryal	Team Project Manager	Avash (Sep 23, 2026 20:59:21 CDT)	09/23/2026
Alok Parajuli	Team Member	Alok Parajuli (Sep 23, 2026 20:57:59 CDT)	09/23/2026
Ashok Paudel	Team Member	Ashok Paudel (Sep 23, 2026 20:56:35 CDT)	09/23/2026
Michelle Su	Team Member	Michelle Su (Sep 23, 2026 20:58:37 CDT)	09/23/2026
David Bendeck Cortes	D2 Project Manager	David Bendeck Cortes (Sep 23, 2026 21:01:30 CDT)	09/23/2026
Dr. Dinusha Herath Mudiyanselage	Faculty Advisor	Dinusha	09/23/2026
Dr. Dinusha Herath Mudiyanselage	Sponsor	Dinusha	09/23/2026
Dr. Mark Welker	Instructor		

Contingencies

This section contains any approval contingencies as noted by the Product Sponsor, Advisor, Mentor, or Instructor.

Approver Name	Contingency
N/A	N/A

Engagement Modifications

This section contains any deviations or additions to the Sponsor, Advisor or Team engagement requirements as listed in the Appendixes.

Appendix Section	Modification
N/A	N/A

Section	Author	Word Count
Executive Summary	Avash Aryal and Michelle Su	300
Table 1: Preliminary Subsystem Responsibilities	Avash Aryal and Michelle Su	N/A
Top Level Block Diagram	Ashok Paudel and Alok Parajuli	207
Figure 1: Top-Level Block Diagram	Ashok Paudel and Alok Parajuli	N/A
Product Feature 1: Solar and Wind Energy Harvesting	Alok Parajuli	199
Table 2: Renewable Source Parameters	Alok Parajuli	N/A
Product Feature 2: GaN-Based Boost Converter with CC/CV Battery Charging	Avash Aryal	150
Product Feature 3: Real-Time Monitoring and Display	Ashok Paudel	148
Product Feature 4: GaN-Based Buck Converter and 5 V Power Regulation	Ashok Paudel and Avash Aryal	127
Product Feature 5: Energy Storage and 5 V Load Supply	Michelle Su	167
Table 3: Milestone Schedule	Ashok Paudel	N/A

6. Appendix

A. Sponsor Engagement

1) Project Team Responsibilities

- The Project Manager will set up and facilitate a weekly call/meeting with the Sponsor. The Project Team will provide weekly status updates to the Sponsor including upcoming deliverables, critical issues, and any adjustments to the Project Plan.
- Documents will be provided to the Sponsor with adequate time for review and signature. The time necessary for review will be agreed with the Sponsor. The minimum review time will be 3 days prior to the document due date.
- Design files will be provided to the Sponsor as requested in a format agreed to with the Sponsor.
- Support requirements will be clearly requested from the Sponsor with the dates required and an adequate time for fulfilling the request.
- The Project Team will respond to Product Requirement, (PRD), change requests from the Sponsor within 1 week with a description, plan, and rationale for accepting, rejecting, or modifying the request.
- Modifications to the Product Requirements, (PRD), requested by the Project Team will be communicated to the Sponsor immediately with a description, plan, and rationale for the request. No changes to the product will be implemented until agreement has been reached with the Sponsor.
- Modifications requests to the Project Plan by the Sponsor or Project Team will be resolved and documented within 1 week of the request.

2) Sponsor Responsibilities

- The Sponsor will participate in a weekly call/meeting with the Project Team to review the project status, upcoming deliverables, priorities, issues, and progress to the agreed Project Plan.
- The Sponsor will provide document review, feedback and approval, rejection, approval with contingencies with adequate time for the Project Team to meet the course due dates. The Sponsor will clearly outline the time necessary for deliverables review. The minimum review time will be 3 days prior to the document due date.
- The Sponsor will provide feedback to requested support requirements from the Project Team.
- The Sponsor will respond to Product Requirement, (PRD), change requests from the Project Team within 1 week with a rationale for accepting, rejecting, or modifying the request.
- Modifications to the Product Requirements, (PRD), requested by the Sponsor, will be communicated to the Project Team immediately with a description and rationale for the request.
- Modifications requests to the Project Plan by the Sponsor or Project Team will be resolved and documented within 1 week of the request. If necessary, as agreed with the Project Team, the Sponsor shall provide technical advice from a subject matter expert(s), to provide guidance to the Project Team answering inquiries approximately 1 hour per week from start to finish of the project.

- Attend one design review during each semester and attend Senior Design Day each semester.

Sponsor Support Elements		
Element	First Needed	Needed Until
Project Materials	09/21/2026	05/10/2027

B. Faculty Advisor Engagement

1) Project Team Responsibilities

- The Project Manager will set up and facilitate a weekly call/meeting with the Faculty Advisor. The Project Team will provide weekly status updates to the Faculty Advisor including upcoming deliverables, critical issues, and any adjustments to the Project Plan.
- Documents will be provided to the Faculty Advisor with adequate time for review and signature. The time necessary for review will be agreed with the Advisor. The minimum review time will be 3 days prior to the document due date.
- Design files will be provided to the Faculty Advisor as requested in a format agreed to with the Advisor.
- Support requirements will be clearly requested from the Faculty Advisor with the dates required and an adequate time for fulfilling the request.
- Modifications requests to the Project Plan by Faculty Advisor will be reviewed and agreed to within 1 week of the request.

2) Faculty Advisor Responsibilities

- The Faculty Advisor will participate in a weekly call/meeting with the Project Team to review the project status, upcoming deliverables, priorities, issues, and progress to the agreed Project Plan.
- The Faculty Advisor will provide document review, feedback and approval, rejection, approval with contingencies with adequate time for the Project Team to meet the course due dates. The Advisor will clearly outline the time necessary for the deliverables review. The minimum review time will be 3 days prior to the document due date.
- The Faculty Advisor will provide feedback to requested support requirements from the Project Team. This includes feedback and guidance on design implementations decisions, design files, test plans, test procedures and test results.
- The Faculty Advisor shall provide technical advice and guidance to the Project Team, answering inquiries approximately 1 hour per week.
- Modifications to the Project Plan by the Project Team will be resolved and documented within 1 week of the request.
- Attend design reviews and presentations each semester providing feedback and guidance for the Project Team
- Attend Senior Design Day each semester.

C. Ground Rules

As a team and as individual team members, we agree to:

1. **Stay focused on our objectives and goals.**
   Each time the team meets, we will clearly define our objectives and desired outcomes at the beginning of the meeting. We will politely remind team members if we are getting off track.
2. **"Sidebar" any issues that are relevant but not consistent with the immediate objectives.**
   Occasionally, important matters are raised that are not relevant to the immediate goals of the meeting. To keep the group on track, but avoid losing the issue, create a "sidebar" where these topics can be listed and discussed later.
3. **Listen when others are speaking.**
   We will listen and consider others' input before adding our own comments.
4. **All viewpoints will have an opportunity to be heard.**
   We understand that some team members may be quieter than others. We will make an effort to get each team member's viewpoint and that no one dominates the discussion.
5. **Differences of opinion will be discussed respectfully**
   We will identify areas of agreement before assessing areas of disagreement. We will encourage each other to look beyond our own point of view. We will discuss different ideas respectfully. As a team, we will weigh the merits of different opinions and agree on a process for choosing a direction. All team members will respect and follow the decision or direction.
6. **Look for the good points in new ideas.**
   We will endeavor to explore the value in each idea as we assess and select our path forward.
7. **Focus on the future, not the past.**
   We will use our experience to inform our decisions, but focus the discussion on the future objectives. Blame for past performance is counterproductive, we will focus on finding solutions.
8. **Agree upon specific action items and next steps.**
   At the end of each meeting and discussion, we will summarize and agree on specific next steps, action items and assignments.
9. **Accountability**
   As team members, we will each be responsible for our individual assignments and contribution to achieving the team objectives and goals. We will honor our responsibilities and not let our team members down.